Khoa Nguyen

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Education

San Jose State University — San Jose, CA

B.S. Computer Science | GPA: 3.92 / 4.00 | Dean's Scholar — *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Design, Databases, Discrete Math.

Experience

SJSU College of Engineering (SJSU SCE) — San Jose, CA

Software Engineer Intern — *Dec. 2025 – May 2026*

- Built event registration for Clark, the SCE website for **1,500+** engineering students, from the React sign-up and admin pages to the Go (Gin) REST endpoints and MongoDB behind them.
- Backfilled missing fields in old MongoDB event records with a Go migration that is safe to rerun, because it only updates records that are missing a field; the team now runs it automatically on the server.
- Gave event admins one paginated sign-up list with pending, accepted, and rejected counts, through a Go (Gin) endpoint limited to that event's admins and a React dashboard on top.
- Kept capacity edits from overbooking events: the Go update endpoint recomputes remaining seats in Redis as the new capacity minus seats already taken, instead of resetting the counter.

University of South Dakota — Vermillion, SD

Member Assistant, University Coding Club — *Aug. 2024 – May 2025*

- Led weekly algorithm sessions for **35+** club members, presenting a worked solution to each week's challenge.
- Mentored peers on debugging, complexity analysis, dynamic programming, and graph traversal.

Projects

Cotune | *Java, Spring Boot, GraphQL, WebSockets, PostgreSQL, Redis, React* — Jul. 2026 – Present

- Built and deployed a real-time beat editor, like Google Docs for music, with Spring Boot and WebSockets; Redis keeps edits in sync when the app runs on more than one server, and a test that runs two servers proves it.
- Fixed editor lag by caching a Spring Security permission check that ran on every cursor move: about **40** database calls a second per user, sharing only **5** connections, went to **0**, and a test now fails if they return.
- Added an AI composer with Gemini function calling that turns a text prompt into a plan of edits; the server checks every edit the same way it checks a person's, and nothing is saved until the user approves the plan.

FluxCareAI | *Python, FastAPI, uAgents, Gemini, Perplexity, Groq* – CruzHacks 2026 — Jan. 2026

- Built the FastAPI backend of a mobile app that turns a product photo into a one-page safety report, with each AI agent running as its own service: Gemini reads the photo, Perplexity finds recalls, and Groq writes the report.
- Kept the app from hanging on slow AI agents: the API waits for the writer agent to send the finished report back to a webhook, and returns a timeout error after 90 seconds instead of waiting forever.
- Kept reports tied to real sources: every search finding keeps its links, near-duplicate findings are dropped, and the writer model is told to use only those findings.

PlaytestPilot | *Python, FastAPI, Docker, C#, Unity* – SJHacks 2026 — Apr. 2026

- Won **1st place** (Game Track) at SJHacks 2026 with a tool that plays and tests Unity games by itself; built its Python FastAPI backend and the agent that runs Unity in the background, on a team of 4.
- Made sure the Unity agent never picks up a half-written job: the FastAPI backend writes each job to a temp file and renames it into the queue only when it is complete.
- Made the Unity agent restart itself after crashes: the backend marks it offline when it stops checking in for 10 seconds, and a Bash script restarts it when the process dies.

Technical Skills

Languages: Java, Go, Python, TypeScript, JavaScript, SQL, Bash, C#
Frameworks: Spring Boot, Spring Security, Gin, FastAPI, React, Next.js, Node.js (Express), Unity
AI: Gemini API, Groq, Perplexity API, uAgents, LangChain
Databases: PostgreSQL, Redis, MongoDB, SQLite
Tools: Git, Docker, GitHub Actions, AWS, Linux, Heroku, Nginx, Prometheus
Testing: JUnit, Testcontainers, Playwright, pytest, Go test

Awards & Honors

1st Place, Game Track – SJHacks Hackathon — 2026
2nd Place – AWS Cloudathon — 2026
Honorable Mention – ICPC North Central Regional Programming Contest — 2024